

Hi everyone and welcome to the lecture on evaluation for the first one of the two weeks of lectures that will look at different kinds of evaluations.

So how can we make sure that our systems are doing what we want to be doing?

So, so far within the lectures as well as the SGTs and LGTs we've been going through this double diamond process.

starting with just trying to understand what the problem is what the users think then defining it down to a specific design brief and then more recently really starting to ideate and play and prototype with some of the examples now we're coming to the end of of the double diamond where we might be starting to look at how do we test how do we evaluate whether the prototypes that you've built are actually useful and you'll see that these arrows that kind of go back into you test and then you usually come back to ideate, then you have to re-prototype because you've discovered something interesting. I expect that that is what's going to be happening as well. So it's not, as always, it's not a linear process. It's something where you might cycle through individual places or maybe even go back to redefining your problem, although that's probably not going to happen for you within the context of this course.

So as I said, this is the first of the two weeks where we will be looking at different kinds of evaluation techniques that you could use for your designs. And the general approach that we will want you to use as always is that really evaluation methods should be thought about as a set of tools that you have at your disposal and which one you choose is really dependent on the kind of questions or assumptions that you're trying to solve. So we'll be talking about how different methods might look depending on your stage of the design or the types of questions that you're asking so across all of these you might be asking why to do evaluation testing it's very much to check that the specific aspects of user experience that you're interested in are considered in the design and are implemented well in the design you might be testing your conceptual models you might be testing some of early prototypes you might be trying to compare your system to other products or just double check if it actually does the work that you want it to do so does it increase critical thinking when people use it for example there's some variation around where you look at it it might be natural settings so meaning you would deploy as part of a class maybe next year's class you might be exploring it in the wild as in you just open it up on on amazon or

apple and you will see how people use your app. You might be thinking about labs or specific lab arrangements.

And also you might be thinking about when throughout the design you deploy that testing. And we've already done that, as you might remember, a couple of times throughout the LGTs, where we've asked you to explain what you've done to other people. That has been part of evaluation, although a very informal and interesting one.

So in the next couple of minutes we will go through this idea of how does a map look like of the types of methods that you might want to know about and that you might want to be using. Each of these has a specific purpose, they might be used in specific situations and we'll talk a little bit about what that looks like and how do you make sense of this map.

We will however focus only on a few of these methods in detail just because we don't have time and I don't want to turn this into another two-hour video with a bunch of reading theory on the side. So what are the kinds of methods that we could look at?

So one potential way of starting to map out how these tools look like, how your evaluation tools look like, is to thinking about whether or not you involve your target users. So in your case probably students

who are engaging with critical thinking for example.

One option is to think about approaches that involve experts which might be easier to get or you might rely on a lot of expertise that they might be bringing.

The other big type of studies is the studies that actually do involve your target population in one way or another.

So just to build up your vocabulary some of the methods that include experts are based around heuristics evaluation. So you might be thinking back to Nielsen's 10 heuristics and kind of thinking about how you get experts to engage with your system according to these heuristics. And that's something we'll talk about a little bit later. You might have some models about how your system should behave, and then you could be running evaluations against that. You might be analyzing what the system does. You're not really involving the users, but you're analyzing the traces of what the users do that helps you understand and then I think cognitive walkthrough we've mentioned earlier in the course where you would have a specific way of getting an expert to step through the the process of your system and you're again trying to understand how and when and where your design potentially breaks on the other hand when you have users the useful differentiation is around where and

how you engagement so this could be in the lab or any sort of controlled setting where you'll be asking users to do things that they might not be doing naturally.

So you're asking them to do controlled tasks and then perhaps observe differences between how they engage with these specifically defined tasks or the tools that you provide them. You could think about deploying these in the field, so that means deploying your prototypes in a relatively messy real-world context where they would end up being used. So in your case, it would be about taking a system and deploying it with people who are actually trying to study for an exam, for example.

And finally, you could also be thinking about doing living laboratories
Timestamp 6 minutes where you're essentially putting participants into instrumented environments for long periods of time, but still with some control and data collection, which I think is going to be less common, especially for the types of problems that we're talking about. Finally, you could also obviously do surveys where you, for example, try and reach out to every single student within KINGS and you ask them something about how they either use an existing system, if you were, for example, trying to evaluate how beautifully designed Keats is, or if you're trying to understand something about their

processes and the challenges
and you're trying to
evaluate the challenges that
they face, for example,
with critical thinking.

So, when we start thinking
about these in more detail,
again, just to build up
vocabulary for you, within
the lab studies, we might
be talking about think
-aloud protocol, where you
get a user to go through your
system and discuss how and
when this works. Sorry,

I'm just realizing my son
is marginally in the way.

You might also be able
to engage in experiments,
which is something that
we'll spend a lot of time
talking about. And you might
also be looking at a specific
type of engagement within
a lab that looks at
usability evaluation, where
you're not really interested
in the effects of the
system, but you're really
interested in understanding
how someone will go
through the system and are
there any barriers to use.

In terms of the field studies,
there are technology probes,
which we'll discuss next week.

You could be looking at
deployment of prototypes
and understanding how
these end up being
used and appropriated,
as well as A-B testing,
which is essentially a
field study experiment or
a particular kind of
experiment that many of
the big tech companies
are running right now.

So if we put this all together,
you can see that really

there's a large number of different things, and we Timestamp 8 minutes haven't, frankly, covered everything, but there's a large number of different tools, each of which has specific purposes and specific reasons for its existence.

So it answers particular kinds of questions pretty well, and it also has some limitations, so it doesn't answer different kinds of questions at all, possibly.

So what we'll be doing over these two weeks is really trying to focus mostly on the methods that are often done a little bit earlier in the process, in the sense that they might be giving you formative information whether the design is on the right trajectory.

So you're often using these methods to sense check and observe before you might commit yourself to your final push or deployment, or it might be just after you've deployed and you really want to understand whether the system has a chance of having the effects that you want it to have.

So, for example, you could imagine you're designing a new version of an Instagram interface or some other critical thinking tool, as Timestamp 9 minutes you do in your LGTs, and you might do a heuristic evaluation on semi-interactive sketches to just double-check whether your user interface is understandable for people. Then you might follow by a think-aloud prototype on a, sorry, a think-aloud protocol that

you would deploy on a fully fledged version of a clickable prototype that you've either bytecoded very quickly or you've done in Envision or Figma and if everyone likes this you might have a rough experimental version that you test out within a lab study as a small experiment to see whether the processes whether the AI scaffolding that you're providing to users is working the right way and potentially eventually send this out as a technology probe where people actually try and use it over a period of a week or two in similar ways that an MVP might be used in a startup.

So across all of these, as you see, there are different types of questions and layers of questions that get emphasised. And what we'll be doing in the rest of this week is to focus on about half of these methods and then fill out the rest in the week after.

So, by the end of the next week, you should be able to start thinking about these methods as being positioned across these two or three different dimensions.

How abstract or concrete these are so are you abstracting out something about an interaction that you want to test but it's not really looking at what people will actually do with the system or how concrete are you asking about the things that people actually do with your system how obtrusive is it so do you

need to get people into
your lab or do you ask them
to fill out some sort of
survey questions outside
as opposed to how unobtrusive
where you might be just
tracking what users do
generally And then we will
be talking about the types
of goals that you might
have. Are you more interested
in realistic use? Are
you more interested in
generalizability? Are you more
interested in precision of
kind of unpacking something
about, for example, the
psychological effects?
So we'll keep on returning
to this little diagram,
which probably doesn't
make too much sense to you
right now. But I think it's
a helpful way of always
sense-checking, are you
able to think about the
methods that we'll be
talking about in these terms?
So, what's the plan for today?
I really want us to
focus on things that you
can apply directly
within your LGT and SGT.
That's coming up. So that's
why we'll be talking a
little bit about heuristic
evaluation and then a little
bit more about ThinkCloud
studies and protocols, because
I think that's what will
make sense for you to start
doing during the LGT. We
will then also spend a lot of
time thinking about what
is the theory behind
experimentation? What does it
actually mean to experiment? Why
do we do it? Why do we care
about understanding the
causes and unpacking whether

your system actually improves,
for example, critical
thinking or something else?

And that will be reasonably
complicated, and we'll
spend quite a bit of time
on this, just because I
think that's one of the
pieces that are really hard
to understand just by
reading a website. If I send
you to a Nielsen's
heuristic evaluation page,
within 10 minutes you'll
have a decent idea of how to
do it. If I send you to
a page about experiments,
it's going to be much
harder to pick up.

And both of these things
are then the grounding
for us to start thinking
about the follow-up studies,
which will be field
studies really thinking
about technology probes
and so on, which are
particularly helpful within
generative AI systems. And
then red teaming methods
that I was talking about,
which is another way of
kind of sense checking
and working either with
experts or your users to
ensure that your system
has a lower likelihood of
breaking in ways that are
particularly difficult.

All right, so
let's get started.

Okay, so let's get
started. with the two
lab study type
evaluations, the heuristic
evaluation and think
-a-lot protocol.

So in heuristic evaluation,
experts guided by a set
of usability principles

that we've talked about
as the heuristics,
evaluate whether user
interface elements, such
as dialogue boxes or
menus or navigation
structure or your chat
boxes, online help, so on,
conform to tried and
tested principles.

So these heuristics
closely resemble high
-level design principles,
so making designs
Timestamp 14 minutes consistent, reducing
memory load, and so on,
making sure that you're
really using terms that users
understand that the mental
models are consistent,
and so on. So as you might
remember from some of the
early weeks, the original
set of heuristics for HCI
evaluation was developed by
Nielsen and his colleagues,
who derived them empirically
from a large number of
usability problems. And
then the revised version of
these heuristics is listed
in the next slide and that's
what we've been talking
about in the earlier weeks.

So when you think about
heuristic evaluation it
is something that's
conducted by experts who
understand these heuristics
and it really just takes
a look at your interface
and tries to verify whether
there are some problems.
So the example here
is we have text that is
white and kind of overlaps
with the product images
and it makes it all
confusing and complicated
and probably hard to rate.

So you should now be able to imagine quite easily what would it mean to go through a system with the focus on, for example, user control and freedom and really start looking for the places where the design that someone has done or that you have done breaks that kind of heuristic. So it might be that you get the user into a state where they have to finish an interaction that might take five minutes and they have no way of coming back out of it. That would be an example of a problem with that particular kind of heuristic. So how does this actually look like if you're trying to run it? The playbook is that you need to go and recruit a set of experts, so people who understand what these heuristics are and have ideally some experience with design and designing systems then you need to focus them on what they're supposed to do so often you will be doing prepared scripts and guides to ensure that everyone kind of understands enough about the user task

Timestamp 16 minutes the specific use task that you have in mind and that you're asking them all to focus on the same set of heuristics for example or the same part support of the system. Then there's something about what the experts actually do. Often they have to take at least two passes through the interface. The first one is just to get a feel for how the interaction works and what the product

does, and then the second pass is really getting them to focus on specific elements and follow your protocol.

And you're identifying these usability problems over time in an iterative

way as you get that data from experts.

And then, in the end, obviously, you provide an overall summary of how the experts engage with the system, what are the issues and what are the recommendations that they came up with.

So, relatively simple, it's kind of a very high level, well, high level, it's a very surface level thing that really tries to make sure that regardless of what the system does behind the scenes or how well it does or does not work, at least you're capturing some of these heuristics and you often will

see it as something that's easier to recruit, especially if, for example, you have these experts within your team were ready, it's often that kind of a first cheap low-cost method for you to engage with an application that you've developed before you have to go out and try and recruit participants.

So the think-aloud method is you could imagine a flipped side of the coin when you're trying to really engage with participants rather than experts and you're trying to get a sense of how they experience the system that you've designed

so the definition is that in the thinking aloud test you're really asking the participants to use the system and you might actually be giving them specific tasks of what they're supposed to do so you know take this lecture file upload it here then try to ask questions that go in this direction and so on so they'll still get some sort of script but what you're asking them to do is provide you with a running commentary, verbalizing in an ongoing way, what is it that they're thinking about as they go through the interface. So it's about, oh, this sounds a little boring, and I don't understand this. And now I'm thinking about what should I type in here. So let me let me try and type in this and so on. So this will be one of the tasks that you'll be asked to do.

You'll be asking other people to do on your systems in the upcoming LGTs.

So what's what's happening in practice is that you're really asking users to talk while they're performing tasks those tasks should be something that you're really interested in so they'll probably be tapping into some of the core features that you have and for the users it's slightly unusual because you have to verbalize what's going on through your head as you're engaging but what you as the designer really want to get is what is it that they're thinking

what is it they're trying to do which might not often be obvious. If, for example, a user stops and is not really engaging with your system because they're trying to figure out what to say or what to write, you wouldn't normally know if they're confused or if they're just thinking quite hard.

You want to hear any questions that arise as they work. And also, what is it that they're reading? Like, which parts of the system are they actually engaging with? So, to really kind of nail that down, you need to know what users are thinking in this type of method, rather than just observing, having a little screencast recording, for example, of what they're doing. And that's what makes the method quite powerful.

So, how could you do this practically?

Timestamp 20 minutes You kind of sit there with the user, as this amazing diagram shows.

You might have evil thoughts, like, oh no, that's not what you're supposed to do. Oh my god, how can you not understand? But you're supposed to be really quiet. The only things that you're allowed to kind of say is, tell me what you're thinking.

You're not supposed to answer their questions. You're just trying to elicit their thought process as if they were engaging with the system on their own.

But you want to make sure that they're verbalizing

in an ongoing way. So this part of only help on things that you have pre-decided is really important. And you need to keep track of what you were helping people with. So it's not a system about how someone can use your system if they have an expert who guides them through it. It's really you trying to get a sense of how people would engage with the system on their own, but you get all this additional layer of information that they capture by talking.

So what you want to do is you want to record what's happening, as well as do some good notes throughout, so we'll have to figure out how to do that at the LGTs, so that you can really get a good sense of what is it that was going on for them, so you'll probably want to capture the screen and you might want to, you know, see, have some watch, but I don't think that's too necessary for our tasks, apart from understanding how long it takes people to go through the individual things you're asking them to do, as well as you want to record audio and video and event blogs and so on.

So really have a good capture of what's going on and how the participant has engaged with your system.

So let me just throw this diagram again at you and show how both of the things that we've talked about so far have been about these lab experiments. And what

we were trying to do is really get at some precision of types of questions that we're interested in. So in both cases, the heuristic evaluation was really looking at answering specific questions about whether experts think that our design breaks some established rules that are kind of instantiated within these heuristics.

And within the think-aloud protocol, we were really interested in what is it precisely that goes through people's heads as they engage with a particular tasks that we've set out for them in a place where they come to us and we need to be there with them, whether online or remote. And we're being quite obstructive,

We're trying to get them to spell out exactly or talk aloud exactly about what's going on in their head.

So it's not particularly realistic. It's not what people would be doing. But what we're buying by a lack of realism is that we're really able to get precisely at the types of questions that we are interested in.

All right, let's now switch gears and start thinking about experiments.

The two methods that we've talked about so far are really just ways of collecting patterns of interest.

So, for example, just because an expert says that this button will confuse users and force them to create errors, it doesn't mean that we've proven that

that observation is actually true or that changing that button will fix anything. Although we might trust their expertise and trust kind of the heuristic patterns that are already embedded in that evaluation method, and think that it might be quite likely.

Similarly, if we notice that three users out of five in our think-aloud protocol really loved our systems and the other two haven't, and these three were also wearing glasses, that might not necessarily mean that glasses are an important predictor of our system being successful and that we have to kind of force everyone who is to use our system to wear glasses.

So experimentation in general is really just a set of tools and approaches to help us make causal inferences. that is increase our likelihood that we can Timestamp 24 minutes control the outcome of situations or know that if we change something, what will happen afterwards. So for example, it's something that would allow us to have trust in phrases such as, if we change that button this way, then 30% fewer users will drop out of our onboarding funnel. So let's have a look at what the common problems are and how should we think about these, how should we think about these approaches?

So this mantra that you've probably heard, correlation

is not causation, is the key reason for why we have to do experiments and why experiments are relatively tricky to set up. So a famous example is that if you look at these two graphs, we might see that as ice creams go, sales go up, so do shark attacks. So we might hypothesize that this is causal, that sharks are attracted to the ice cream and therefore there's an increase in shark attacks on people, of course, that would be making a wrong conclusion. You might make the same point about sunburn and ice cream. It seems that every time I eat ice cream, I get sunburned. Or this XKCD comics where someone is standing on a chair holding a pineapple because it appears to make their mobile signal better. But is it really? So this notion of just because two things correlate with each other, happen next to each other, or just after each other doesn't mean that they're causally related. Experiments are ways of trying to separate out these kind of accidental correlations where two things cluster from causal correlations. So maybe it is exactly ice cream sales that drive shark attacks because there's more people in the water which brings sharks closer and therefore we get more attacks. We don't know. And experiments are the ways of trying to unpack that if-something-then-something-else types of claims.

Timestamp 26 minutes So, enough of the motivations and the problems.

Let's start looking at what experiments are, and let's start doing that by looking at definitions, which can be often a nice starting point.

And as you will probably know, experiments and this approach to cause identification is essentially a key part of what you would call a traditional scientific method. So in those places you start by stating a clear testable hypothesis. So a hypothesis about if something happens then these other things will happen afterwards. By testable we mean that the hypothesis must be quantifiable and measurable.

So you have to be able to say if this happens then this amount of change will happen. And the thing that we're trying to really measure is the cost. So the effect of, if I shift this starting point, then something else will shift later.

And experiments are essentially a way of trying to get at that cost. So let's have

Timestamp 27 minutes a look at how that works in practice.

So what we're going to do as a little toy example is to start with this relatively simple hypothesis that we might want to test, and that is that the Mac OS menu bar, which is anchored to the top of the screen for those who don't have Macs, is faster to access for users than a windows menu bar which is attached to any

of the application windows
and therefore might be
just all over your place all
over your screen depending
on how big your window is
where it's positioned and
so on so what does that
mean what what what is the
cause we're inferring here
what is it causing how does
it fit into the the theory of
causation and experimentation
that we've talked about
here on the previous
slide so in this instance
what we're saying is that
there's something about the
design decision on Mac, which
we're saying is about where
the bar is placed, that
makes the bar work
differently for users than on
Windows. So in other words,
if we design the bar in the
same way as Windows does,
we expect the users to be
slower to engage with it
than they would be if the bar
was on the top of the
screen. Does that make sense?

So the underlying idea is
that when you're thinking
about experimental
design, you're talking
about these independent
variables, these sort of
things that you expect
are somewhat related to
the cause. And you're
trying to manipulate these
and understand how that
changes some output,
which is often described
as a dependent variable.

So what you can be thinking
about is that the process
that you're trying to
understand, in this case, how
quickly people can engage
with the menu bar, is a black

box with lots of inputs and few outputs. So the black box might be the user using your system with and the controlled experiment twiddles some of the input knobs so for example where the menu bar is stored um so those are the independent variables and then you observe some of the outputs the dependent variables in this case how fast people can access it to see how they're Timestamp 29 minutes affected so for the independent variable we might be saying mac and pc but we could also be saying um you know the the bar position at a particular part of the screen or anything else that we're interested in.

Now the problem often is that there are many other input knobs as well, these unknown uncontrolled variables that might affect the process that we're observing in an unpredictable ways. So often the purpose of running an experiment is to really eliminate the effects of these unknown variables or at least render them kind of distributed across both of the conditions so that we can really draw conclusions about how independent variables affect the dependent variable. So that's the epsilon at the bottom here.

So what could some of these unknown variables be? If we are still thinking about the menu bar experiment, some factors might affect how fast the user can reach the menu bar. So for example, the pointing device that

they're using, whether it's a mouse or jackball or joystick or something that they're not really used to, how fatigued the user is, how used are they to one kind of menu bar or the other. So if you have someone who's spent their whole life on Windows, they will be looking for the menu bar at the top of the screen and so on.

So all of these unknown variables might really affect the dependent variable, which is the speed of access to the menu.

So concretely, if we bring in a bunch of people who may have never used a Mac before, it might be that these people are faster on Windows when in reality the Mac, on average, might be better and so on.

So what this means is that in our controlled studies, we're always trying to minimize or at least control and be cognizant of these variables so we can consider them in our analysis.

Which is why experimental tasks are either often more abstract, so it is about I need you to access the menu bar at 10 times in a row, and I'll be varying the position of the application screen on the big screen, for example.

And we also have to be very careful about who we recruit and what are the unknown and uncontrolled variables that we're introducing.

So how would this abstract babble actually work in

our example? So we're saying, as a reminder, we're testing whether Mac bar is faster to access than Windows menu bar. We've already said that one of our independent variables is exactly the cause that we're trying to look at, which is the difference in interface style. So where does the menu bar sit? Is it where Mac puts it or Windows puts it? The other independent variable might be the training that we give to users so that they know where to expect that menu bar. that might have strong effects. If someone's not trained, then on the first time they're engaging with this interface, they might take much longer, not because it takes longer time to move the mouse to the right. Timestamp 32 minutes place, but because they just don't know where to find it, for example. So there's multiple independent variables that we could bring in, depending on what we're interested in. And then obviously, we're measuring some dependent variables. So we are talking about the length of time that it takes. Perhaps we could be looking at errors, we could be looking at satisfaction, we could be looking at how many tasks people actually manage to get done within a time. So all of these are specific instantiation of this hypothesis that Mac menu bar is faster or better than Windows menu bar.

And we're just saying the dependent variables are examples of what the faster better actually means. And then often you will use statistical tests to accept and reject the hypothesis based on your empirical data, which again would normally be a whole course. And I hope that you've had enough statistics in the rest of your classes that you already know what this means. So the important thing for HCI here is to really understand what experiments are trying to do, that we're trying to show that by moving something about our system, something else has a tendency to happen and then really be able to talk about what is the difference between an independent and a dependent variable why is it important to control the other interfering inputs if we want to understand something about causation and so on so we don't expect you to be experts in experiments and experimentation from this very short lecture but it's really important for you to understand at least these basics as we go forward in thinking about what does it mean to evaluate your systems. So to finish, I'll point you to this Campbell and Kronbach chapter that you have on Keats. It really gives probably the best overview of the ideas behind experimentation, what it means, and how to run experiments that I could find throughout the literature. It doesn't really just explain what

experiments are, but there's a lot of deep kind of discussion about why experiments work the way they do.

So I think it's really important for you to go through that text and understand quite deeply what the claims we're making when we are running studies and what are the things that we need to avoid. I'm conscious that the document, you know, that the PDF might be quite difficult to read but it will really help you understand this stuff in much more depth and at your pace.

There's also interesting bits in there about the philosophy of science that kind of helps us understand this movement from the value of local observations, meaning, you know, we interview a couple of people and say, ah, 10 people out of 15 have had this problem, so we should probably change our design, and then contrast these kind of approaches to what we've recently discussed, which is talking about causation and generalizable statistical patterns, And what does this mean? So, for example, when you're wanting to make claims such as a design of a generative AI tutor will lead to 30% higher learning if we include feature X rather than feature Y, you do need to understand causality and you do need to understand experiments.

So let's quickly get into some of the threats

that experiments face.
And these are often
reported or talked
about as the threats to
validity and reliability
of your experiment.

So the three top problems
that cause threats
to reliability and
validity are internal
validity, external
validity, and reliability.

And I'll go through each
one of these at a time.

So internal validity refers
to whether the effects
we see on the experiment
outputs, so for example
how fast people have moved
to the menu bar, were
actually caused by the
changes we made to the
inputs. So for example
whether we gave them the Mac
version or the Windows
version. or if they were
caused by some other unknown
variable that we didn't
control or measure and
we know nothing about.

So if you take the
comic on the right,
the boy might assume that
there's a cause between
the seatbelt sign turning
on, which often happens
before you hit turbulence,
and the plane starting
to move. So from his
perspective, he's saying,
well, my experiment is
anytime I see the sign being
turned on, the plane gets
bumpy, and that's not great.

So statisticians call
this effect, so the sign
gets turned on only when
another variable interjects,
which is you're flying
into a turbulence.

So statisticians call this effect confounding.

When a variable does not, that we didn't control, has a systematic effect on the dependent variables.

Thinking back to the Mac and Windows bar experiment, if we accidentally only recruited long-term Mac users and then collected only one to two trials with each user, it would be very likely that the Mac interface would win whatever it is that we would be measuring, but that's less about how the design operates and more about what our users used to and their skill on operating Mac and just general hatred of Windows that might then lead to the impact.

So it's the knowledge of the interface that the users are bringing in, which would be the uncontrolled confounding variable here.

The cost of this approach is external validity. So if you control too many things, then you might turn the experiment into something that's so focused that it doesn't resemble the real tasks in the world outside of the lab anymore.

So when this happens, for example, if we try to control all the factors that might affect the speed and menu access, that would mean that we fix the starting mouse position, we fixed the menu bar, there are only specific choices that don't necessarily make sense, we fixed exactly the

hardware and had one particular user to this, it would be very hard to argue that our lab experiment Timestamp 38 minutes then generalizes to how menus are used in the varying conditions that are encountered in the real world.

So that trade-off between internal validity and external validity is always kind of a big question in study design and it's something that you need to compromise, kind of find a good balance in between.

reliability refers to whether the effect we see, so the relationship between the independent and dependent variables, such as the Mac bar position and Windows position and how fast a mouse can get there, is repeatable. So if we ran this experiment again, would we see the same effect or would we see something totally different?

If our experiment involved only a single trial, so clicking on the menu board just once, then even if we had constant every other variable we could think of, unknown variations such as how focused the person will have been at that specific time will still cause some extra error in the experiment and single Timestamp 39 minutes data sample is rarely a reliable evidence. There will always be some random variation in people and the way how they do things, which means that often for reliability, you will be repeating a lot of things.

Okay, so let's very quickly have a look at typical dangers that happen to internal validity in HCI -style experiments and some solutions that we have.

So you'll notice that very often those solutions tend to come in two flavors.

Either you randomize something so that you kind of prevent the intervening variables to have systematic effects on your dependent variables, or you try and control and kind of hold constant things that could have effect.

So the first big thing are ordering effects. People are annoying, they learn, they get tired. So if you give people 10 similar tasks, I don't know, adding five digit numbers, and you change interface in between the tasks, if you keep the same order of the interfaces, probably the fifth interface will always turn out to be slowest, not because it's worst but because people will have become really tired or annoyed and they don't really want to do that kind of addition anymore.

So often a good way is to not present the task interfaces in the same order and you kind of randomize or counterbalance the ordering in between participants. The other piece are around selection effects so very often humans will be annoying that they come in with specific skills. So for example, when we talked about this hypothetical idea of recruiting

lots of Mac users and no Windows users, and then assigning them to Windows and Mac conditions, the Mac will always win because people are used to this. So the idea is that you want to think carefully about what are the types of populations that would have impact on your experiment ideally not sample your users only from specific sub populations and then when you do sample users again you Timestamp 41 minutes try to randomly assign them to independent variables.

Additional threats to experimental validity are about experimental bias so if you've spent your half a year building an amazing project and now you're introducing that to a participant and you're then asking the participant to compare it to some sort of competitor's app, you might be much more excited about your own project than the competitor's and then maybe the users will pick it up and behave much nicer or say they much like your product much more than the competitor's one, not because it's better but because they've been kind of biased by the experiment. So the best way to control this is just to be aware of it and make sure that if you are introducing, you're trying to be as unbiased as possible. Ideally, you might not know which of the categories or apps the participants have been randomized into and so on.

Providing equivalent training is also really important for everyone. similarly some of the threats to external validity come up um you know in studies very often so kind of only focusing on people from your class for example will not be a great um population if you're trying to speak about um young people in the uk in general so if you're just sampling from university and from an informatics course there are some obvious threats to you guys are amazing but you're not necessarily exactly representative of the same city of of the whole uk population so those are population threats there might be ecological threats again as we were talking about if you make the conditions of the test that you're getting your participants to do or the task that you're doing, getting your participants to do unrealistic, then very often you might get data from an experiment that is not useful at all. So for example, if you ask people to create an essay that they really don't care about, they won't have any value from whether they do it well or badly, and you're using that task that will take, you know, 40 minutes of time that you're not paying people for to test how well your critical thinking app works well then you will have probably some

ecological validity threats
because people just won't
pay enough attention and
they won't behave they
won't engage with your app
in the way that they would
in the real world again
ideas about training and
and the types of tasks
are all pointers to
potential issues that your
experiment might run into
so in this final last section
i really wanted to talk
about some basics of
experimental design techniques
so we'll run through these
very very quickly and
again the text that you have
in keats is something you
should look into because it
has much more information
about the details but
the two main designs that
we often consider in hci
or a human ai collaboration
are the between and within
subject designs so let's
talk a little bit about
the between subject design
what happens there is that
users are divided into you
know n categories often two
or maybe potentially more
if you have three different
apps that you want to
engage with and then each
of the groups only sees one
of the options so if we
stay with two interfaces a
and b one group of people
will only see an interface
a and the other group of
people people will only see
an interface B and again
you can imagine people in
a are seeing Mac menu bar
people in B are seeing the
Windows menu bar and then
what you're measuring is

how quickly are they able to find things in the taskbar so what you're doing is you're comparing often the mean of the number of seconds it took people to do a task X in the group A and you compare that in the group B. So what this does is that it eliminates variation due to ordering effects. Users can't really learn from one interface to the other, but as you can imagine, what it keeps is the difference between people. So if you have very high differences between people, then someone who really likes Windows and you will have that been randomized into the windows condition might be much faster than if they were randomized into the condition with the mac in contrast the within subject design tries to address that problem of uncontrolled variation in a different way which is that each user then sees both of those interfaces in ideally a random or balanced order what this means is that the results then are compared within each user. So what you're looking at is not necessarily how long it took people to go from X to Y on a Mac versus a Windows machine, but you're interested in seeing whether people were generally faster on Mac than they were on Windows. The problem here is that people can use, people can learn. So while you're eliminating variation due to

user differences, because the user is only compared with self, you are not really dealing with the problem of maybe there's some bleed through from task A to task B from one person to within that one person.

So just to visualize this on some sort of more of a diagrammatic type of interface, the between subject design, as you can see, has individual subjects. they all go through the same pretest and posttest, but some subjects only Timestamp 47 minutes see treatment A, which might be Mac, some subjects see treatment B, which might be Windows, and some subjects see a controlled treatment, which might be working on their phone, for example.

In contrast, the within subjects, you will see again have the same pretest and posttest potentially, but then everyone goes through the three conditions at the same time, having some sort of post-test for that condition after each time. Notice this green kind of part where we're counterbalancing, where half of the participants first engage with Mac and then they engage with Windows, whereas the first half of the participants might have engaged with Windows first and Mac later to kind of balance the learning, as we've talked about it before.

So as a summary, when you're thinking about these experiments, there are pros and cons that depend on what you want to do. The main advantages of the between subjects is that we have no training effects, and there's also potentially less strain on our individual participants, as these might often be shorter studies.

The main advantages of within subjects is that we need fewer participants.

And this is particularly important when you begin to imply some inferential statistics where we need to see a definitive effect and not just a simple comparison of means.

It's also quite nice because we might be able to look at more precisely at the difference in results for the same participant, especially if you expect that there's going to be a fairly large between participant variation in how they engage.

Now, finally, the problem here is that these designs become really complex very quickly the moment you start adding conditions.

So in summary, both of these options are different compromises that try to unpack that difficulty of how do you get
Timestamp 49 minutes at understanding a cause.
If I change A, how will my dependent variable change,

knowing that there's so many other inputs that we don't control within the system.

So just to summarize what we've started talking about today, and I'm conscious that it's been quite long and quite theoretical and much heavier than the last two weeks.

But we have started talking about a fairly in-depth and intricate piece of HCI, which means that there's lots of research methods that include various options, such as the heuristic or think-aloud studies we've talked about today, experiments, field studies that we'll talk about next week, as well as surveys, which I expect that you know.

What I really want you to remember from today is both what controlled experiments are trying to do and why manipulating independent variables and measuring dependent variables is one of the ways in which they do try to get at causation.

Also, I'd like you to think about or remember what it means to defend against the threats to internal, external validity and reliability and how that kind of impacts this experimental manipulation.

as well as being kind of aware of the words of randomization, counterbalancing, and how these are often used to address some of these threats to validity and reliability.

So the very final slide for

this week is again highlighting this notion of we have multiple methods that we can think about and each of these methods has particular advantages and particular limitations that we often need to make compromises about and that we pick based on what we really are interested in and what we're willing to kind of not focus on as much. And that's often why research is also triangulate. So you might run an experiment, but then you might also run an observational Timestamp 51 minutes study or a think-aloud study to understand a little more about why and how people are engaging with your system combine the results from multiple methods. So with this I'll leave you to your reading and I look forward to seeing you at the LGTs and next week we'll be talking about red teaming and we'll also be talking about technology probes as the final two methods that you'll need for this course. See you then!